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Baseball

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Performance Demands of Baseball

In this chapter, we turn our focus to the sport of baseball. Beginning with the physical demands of the game that are required for success, we break down the game into its physiologic targets for practitioners, including factors that lead to sustained health and performance throughout a season, and importantly, how high-intensity interval training (HIIT) can be used to help facilitate these outcomes.

Sport Description and Factors of Winning

Baseball is a game played with a bat and a ball between two teams of nine players. The teams take turns batting the ball (offense) and fielding the ball (defense). The first professional league for the sport formed in the United States in the late 1800s, and growth of the sport has been promoted globally through the Little League World Series and the World Baseball Classic. Today it is one of the most popular sports in the United States, where as just one example, it is known that over 40 million viewers watched the seventh game final of the 2016 World Series.

The game has a fairly consistent format globally with the primary adjustment being in the number of innings. The inning is the period of play during which

each team alternates between offense and defense (3 outs). Typically the game lasts nine innings and about 3 h, however, it is often adjusted for youth games. During the Major League Baseball season, currently 162 games are played per team over 180 days, which is preceded by 27 preseason games played over the course of 29 d. An additional 20 games can be added over 30 d during the end-of-season playoffs.

The game is played on a field with two foul lines that intersect (90 degrees) at home plate. The foul lines serve as the primary boundaries of the game. There are four total bases that are 90 ft (27.4 m) apart from each other. Additional dimensions of the baseball field are illustrated in figure 22.1. The dimensions provided are for professional games, and typically these distances are scaled for youth games. The field expands from home plate forward with no official boundary limiting forward progress, however, there can be a secondary boundary in the outfield, which is required to be a minimum of 325 ft (99.1 m) from home plate on the sides and 400 ft (121.9 m) from home plate in the middle for Major League Baseball. This secondary boundary serves to designate when a home run has been hit (figure 22.1).

Players are classified into four groups depending on their location on the field. As illustrated in figure 22.2, these include outfielders, infielders, pitchers, and catchers, with the positional names indicated. Pitchers start every pitch by throwing the baseball

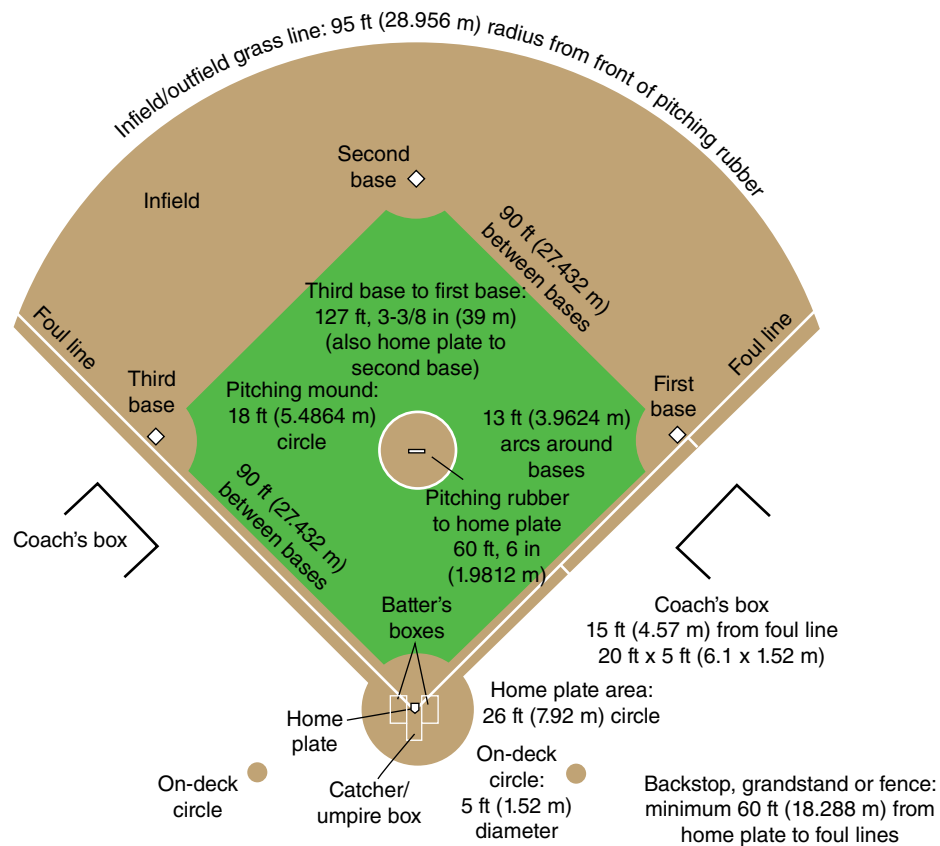


Figure 22.1 Baseball field, its dimensions, boundaries, and bases.

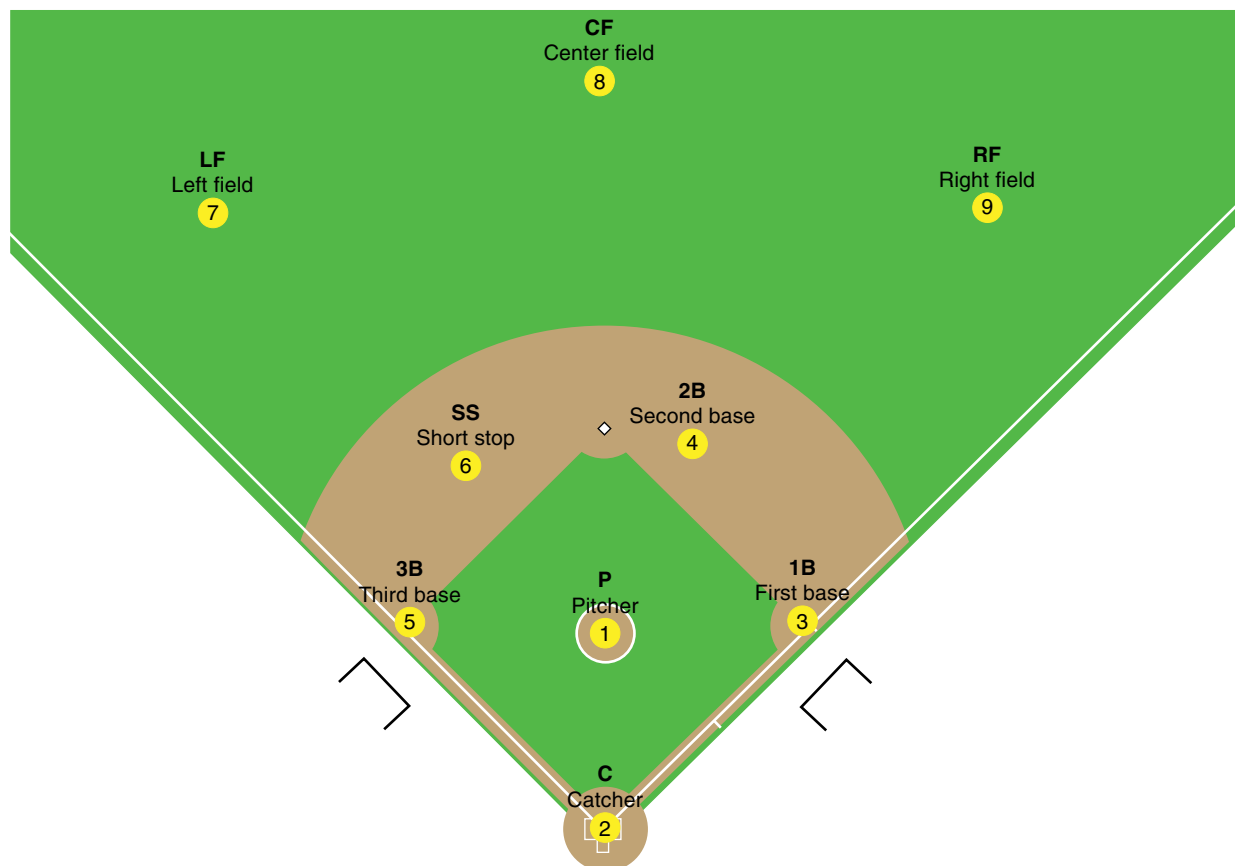


Figure 22.2 General location of the starting position of baseball player positions during most defensive situations.

from the pitching rubber to the catcher who is situated behind home plate. Traditionally the first and second basemen spread out between the space between first and second base while the shortstop and third baseman spread out in the space between second and third base (figure 22.2). There are three outfielders—left, center, and right—who typically start at a location 75% between the bases and the outfield wall in their respective areas.

When the pitcher throws the ball, a number of outcomes can occur in the game, including:

- pitch is caught by the catcher and called either a strike or a ball
- ball is hit in the air in play
- ball is hit in the air foul
- ball is hit on the ground in play
- ball is hit on the ground foul

If no contact is made between the bat and the ball during the pitching sequence, then one of two options occur: a strikeout or a walk. A strikeout occurs if three strikes are thrown by the pitcher and caught by the catcher. A strike can occur by a pitcher throwing the ball in the strike zone and not being swung at by the batter or by the batter swinging at the ball and not making contact. A walk occurs if the pitcher has four throws out of the strike zone that are not swung at by the batter. If the pitcher throws four balls before the ball is hit or three strikes are thrown, then the batter is allowed to progress to first base, or the batter is deemed “out,” before the next batter comes up to bat.

If the ball is hit in the air, the fielding players can run to catch it before it hits the ground in order to record an out. If a ground ball is hit, or if a ball that is hit in the air hits the ground in fair territory before being caught, then the batter runs, without the bat, to first base and attempts to touch the base before the fielding team can gather the ball and throw it to a defensive player standing on first base. If the ball is caught at first base, or if the batter is tagged out with the ball, then the batter is out. If the batter reaches first base, or a subsequent base, prior to the ball being caught by a player touching the base, then the batter is deemed safe and the hitter remains on the base as a baserunner. During subsequent batters, the baserunners look to navigate the bases from first to second to third base to home plate in order to score a run before three outs have been recorded by the fielding team. If a ball is hit in the air that lands behind the secondary boundary, then the batter has

hit a home run and the batter is to run the bases in order along with any other baserunner who was on the bases at the time the ball was hit.

A team is at bat until recording three outs, at which point the teams switch roles to complete the inning. Full games are completed after playing nine innings. The winner of the game is determined as the team that has scored the highest number of runs at the end of nine innings. Games may go into additional innings if the teams are tied at the end of the game.

Success in the game arises from a combination of successful pitching, hitting, fielding, and situational outcomes. Particular outcomes and related factors will be described. The beginning point of each outcome starts with the ball that is thrown by the pitcher. The spin, speed, and location of pitches vary based on the pitcher’s skill set as well as the tendencies of the batter he is facing. The goal is for the ball to be thrown within the strike zone (the height of the strike zone runs from the middle of the chest to just below the knees and is the width of home plate) to the catcher’s glove without the batter making contact. As a result, different types of pitches with various spin patterns and speeds are used to deceive a batter as to the type of pitch being thrown to minimize the chance of being hit. If contact with the ball is made, then the goal is to keep the ball on the ground so that it has a chance to be gathered by the infielders and thrown to first base before the batter reaches the base so that an out can be recorded. To improve the chance of fielding a ground ball, adjustments are sometimes made to align the infield (known as shifting) to improve the chances of catching a ball that is hit by batters who have a greater tendency to hit the ball in specific directions. Outfielders can also be repositioned to improve the chances of catching the ball or due to situational concerns (i.e., keeping the ball in front of them). In order to improve a batter’s chances to make contact with the ball, batters can vary their swing velocity as well as adjust their swing plane related to the type of pitch and the speed at which the pitch is thrown in order to maximize the chance for contact. The goal of batting typically is to hit the ball as hard as possible into a location in fair territory where a fielding player is not able to reach the ball prior to the ball hitting the ground. Once hitters make contact with the ball, they run as fast as possible around the bases until they are tagged out by an infielder holding a ball, stop at a base to prevent being tagged out, or are out at first if the ball is caught at first base by a fielder before the batter reaches first base.

Competent skill execution is imperative to game-related outcomes. The skill of pitching requires fine motor control, endurance, tactical awareness, and power to provide accurate and controlled outcomes that are coordinated with the rest of the players on the team. The average 90 mph (145 km/h) fastball takes 0.4 s to reach home plate, leaving little time for the batter and catcher to react to it. It is estimated that it takes 0.25 s to see the ball and react, which leaves 0.15 s for the motor plan to be executed regarding the type of swing to take or whether to swing at all. When the ball is struck, the average exit velocity that is produced is 86.5 mph (139.2 km/h) (MLB), giving the pitcher less than 0.5 s to react to the ball if hit back at him. As a result, reaction time is an important component of successful skill application for hitters and pitchers. In addition to reaction time, the skill of hitting requires hand-eye coordination, fine and gross motor control, rotational power, tactical awareness, and pattern recognition. When contact is made, the hitter also requires linear and curvilinear speed in navigating the base paths. Agility and change-of-direction ability are required for fielders, pitchers, and hitters under varying circumstances and greatly depend on where the ball is hit. In addition to these

components, long-distance sprint ability is required for players in the outfield most often and also required by batters when balls are hit into play that allow for the hitter to advance to more than one base.

Physical Performance Contributors to Winning

Positive outcomes in baseball require a number of physical factors that span from fundamental aerobic endurance to high-level power outputs during situation-specific fielding, throwing, pitching, batting, and running. These high-level activities account for approximately 10% of the 3 h game. As a result, quick reaction times and powerful movements producing speed are a hallmark of success in the game. Figure 22.3 shows the physical requirements of positional players and the pitcher in baseball relative to other sport athletes. Between these rapid movements, there are extended periods during which focus and awareness are consistently required so that an optimal outcome can occur if a ball is hit into play. At the extreme scenario, the longest baseball game of all time took 25 innings and lasted 8 h and 6 min. With only 25 players available for a game

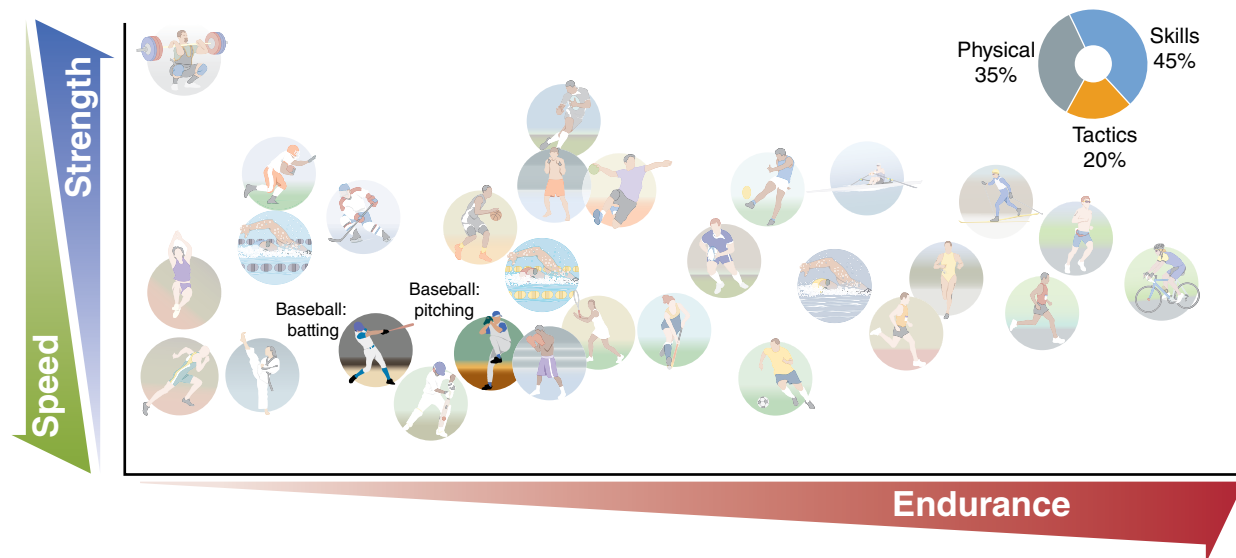


Figure 22.3 The position of baseball players (pitcher and batter) on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging this varies slightly between positions. Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (45%), tactical awareness (20%), and physical capacities (35%) for baseball success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

(roughly 13 position players and 12 pitchers), the physical toll of the game can be extreme, with little time for recovery during the season (6.3 games per wk on average) and consistent travel expectations occurring every 3 or 4 d when a series of games is played away from the home stadium. The game density in professional baseball in the United States is like few other professional sports in this regard. In order to manage fatigue across the season, a high level of aerobic fitness is beneficial for minimizing the effect that the frequent playing has on performance and injury risk (7).

Targets of Physical Performance of Baseball

Due to the high game density and length of the professional season, examining the physical performance abilities of the players is important for maximizing the number of games played and in-game physical performance. Tests of grip strength, body fat percentage, vertical jump height, 10 yd (9.1 m) sprint ability, and repeated aerobic efforts using the 300 yd (274.3 m) shuttle are relevant markers used to create models that identify physical performance attributes of baseball players. Estimates of $\dot{V}O_2\text{max}$ in pitchers have been reported as being $48.1 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ with typical body fat percentages of $15.3\% \pm 3.5\%$ across all baseball positions (4, 5). Across a team of professional baseball players (pitchers and position players), similar estimates of body fat percentage ($13.8\% \pm 3.0\%$) have been reported, while seated grip strength ($111.0 \pm 16.0 \text{ kg}$), vertical jump height ($71.9 \pm 8.2 \text{ cm}$), and 10 yd (9.1 m) sprint time ($1.52 \pm 0.10 \text{ s}$) have also been recorded (5). Three-hundred-yard shuttle times tend to be around 53 s (8).

Hitters and Baserunners

The physical act of hitting during a baseball game is expected to occur within 0.15 s once the pitch has been identified as a pitch that should elicit a swing response. An effective swing requires an efficient transmission of the stabilization force from the back leg to powerful pelvic, trunk, arm, and hand rotational accelerations. During this event, the fingers must grip the bat to allow for efficient energy transmission as well as for adapting to the location where the ball was thrown by the pitcher.

Once on the bases, baserunners require quick lateral agility as well as efficient acceleration to run over a short distance. Typical runs between bases

are 90 ft (27.4 m) long, however the majority of these runs begin 10% to 12% toward the base as players take a leadoff from their current base while the pitcher is completing the throwing motion. If the flight of the ball allows for the players to run to extra bases, then the runner typically completes this skill utilizing a curvilinear running approach to maximize speed around the base paths.

Because of the demands of hitting, it is beneficial for baseball players to exhibit competent lower-extremity stability, lower-extremity power, grip strength, and rotational power of the pelvis and trunk. Prior research in this area has shown that position players (hitters) in baseball display 10 yd (9.1 m) sprint values of $1.56 \pm 0.08 \text{ s}$, grip strength values of $108.3 \pm 13.1 \text{ kg}$, vertical jump values of $70.4 \pm 7.7 \text{ cm}$, and pro agility times of $4.48 \pm 0.58 \text{ s}$ (5). Performance in the 10 yd (9.1 m) sprint and grip strength tests appear to be two physical test areas that are positively associated with higher levels of performance in a cohort of professional baseball players from the United States (9). Differential workload demands likely exist based on at-bat results (hit versus out, etc.) that are made and the power output while on the bases, however, little published research has been completed in this area as of yet.

The majority of the benefits of HIIT training for hitters come in the form of supporting the baserunning activity. HIIT can be used to build and maintain the aerobic conditioning that is essential for health and recovery over the course of a season, while it can also provide the stimulus needed for an athlete to develop explosive power, sprint at full intensity for 20 to 30 s, and recover quickly between efforts. The health and recovery base derived from this conditioning is also presumed to support the hitter in his quest to be able to focus on the act of hitting, as opposed to being distracted from this sporting act due to lingering fatigue or soreness.

Pitchers

Each pitching event is completed in approximately 1 s from the beginning of the motion until the ball leaves the pitcher's hand. The resulting pitch has an average speed of $\sim 92.0 \text{ mph}$ ($\sim 148.0 \text{ km/h}$), and a starting pitcher will typically complete 100 pitches per start. Pitchers who come in to serve in a relief role typically throw 10 to 25 pitches at speeds up to 104 mph (167.4 km/h).

Similar to the hitting motion, the pitching motion requires full kinetic chain involvement in order to

produce the consistent repetitive power necessary for successful outcomes. Pitcher peak heart rates during a game have been reported to be as high as 86.6% HR_{max}, and these peak heart rates typically increase during the early innings and then display a gradual reduction over later innings (82.3% HR_{max} in sixth inning) (2). Grip strength, lower-extremity power, and arm rotational velocity are important factors to consider in supporting successful pitching performance. Previous research has established that grip strength in pitchers tends to be similar to position players, and these values are not observed to change with age (8). However, vertical jump performance in pitchers tends to be consistently lower than position players across age groups, and pitchers observe a greater decrease with these values as they age; this change is not as large in the position players (8). An additional factor that appears to be associated with sustained musculoskeletal health is time between pitches, since it has been identified that longer periods between pitches improves recovery compared to shorter time between pitches (14). While this may be beneficial from a health standpoint, there is a potential tactical benefit of pitching faster, as well as league determinants of how much time can occur between pitches; therefore, this may be a moot point. A number of studies have also established that shoulder range of motion measure in pitchers is an important area to remain normalized in order to reduce the risk of injury to the shoulder and elbow (1, 13).

Compared to other positions, the physical demand on pitchers is the highest on the team. This workload density is the primary reason starting pitchers are on a five-player rotation to allow adequate recovery time between games. HIIT supports the development of qualities important for pitching, as pitchers are expected to produce repetitive powerful efforts (pitches) with relatively short recovery windows between successive pitches. A starting pitcher may throw 150 pitches or more over a game, including warm-up pitches before the game and between innings. Each pitch in competition must be made with similar intensity for the athlete to be successful, and the recovery period between pitches is typically 15 to 20 s. HIIT serves as a valuable conditioning tool, as properly formatted training allows pitchers to build an aerobic base, which is believed to support recovery within and between pitching appearances. HIIT sessions can be an effective means for pitchers to maintain a training effect over the course of a season, while allowing perfor-

mance staff to monitor and modulate load and neuromuscular strain. In addition, HIIT creates more robust alactic and glycolytic systems, thus fueling quick recovery between pitches as well as power production in subsequent efforts.

Catchers

The catcher position is unique in its demands due to the sustained amount of time that the player is expected to hold a crouched position while still being mobile. From this position, the catcher is expected to catch the ball thrown by the pitcher, which is often traveling in a nonlinear direction and may bounce off the ground before being caught. If the catcher does not secure the ball when it is thrown, any hitter on the bases (now referred to as a baserunner) can advance to additional bases. Baserunners can also advance to other bases by stealing bases during the pitching motion. The catcher can get the runner out if he is able to receive the ball and throw it to the infield player covering the base the player is trying to steal in time for the infield player to catch the ball and tag the runner with his glove before the player touches the base. In this scenario, the catcher is required to throw the ball from the crouched position or by quickly transitioning to an upright posture (kneeling or standing) to complete a throw to the base typically within 2 s of receiving the ball. Because of these demands, catchers have specific needs related to the different postures and body motions. Based on these throwing demands, similar attributes to those that pitchers require in a posture-specific setting are relevant to assess. Due to the posture-specific nature of this position, an additional area of assessment for catchers includes thoracic and cervical spine stability as well as mobility in order to ensure that these areas can efficiently adapt to changes in the trajectory of the ball. In addition, lateral mobility and agility in the crouched position are also relevant in order to effectively catch all of the balls thrown by the pitcher. Research is lacking concerning independently examined catcher-specific metrics, with these players typically grouped as position players. Based on the needs analysis outlined, future efforts should focus on these players due to the specific demands of the position.

The benefit of HIIT for catchers is unique compared to other types of players, due to the demands of the position. Catching is one of the most physically demanding positions on the field, as the

catcher is often in a squatting position for 80% to 85% of the time the team is on defense, interspersed with intermittent standing, kneeling, walking, and running over the course of a game. Catchers also typically mirror the same numbers of throws over the course of a game as the entire roster of pitchers used in the game, although each throw is performed at a much lower intensity. In addition, catchers are asked to perform the same offensive roles as other positional players on the field and thus must be prepared to meet the demands of hitting and baserunning. HIIT allows catchers to develop their aerobic and neuromuscular endurance, as well as alactic and glycolytic system power necessary for performance. Similar to pitchers, HIIT is also considered a valuable tool in managing load and neuromuscular strain in baseball players, by facilitating recovery over the course of a season.

Position Players

The infield (four players) and outfield (three players) position players have more physical demand commonalities compared to pitchers and catchers. The primary physical demand difference is due to the distance between home plate and the range of space that the players need to cover.

Infield players typically have a starting position 100 to 155 ft (30.5 to 47.2 m) away from home plate, while outfield players typically start 275 to 350 ft (83.8 to 106.7 m) away from home plate, depending on the position, game situation, and the hitting tendencies of the batter. As a result, the reaction time of the infielders needs to be faster than that of the outfielders in order to stop the ball from getting past them and reaching the outfield. Infield players typically stop these balls by quickly moving or diving laterally, which highlights the importance of lateral agility and power for infielders.

The other factor that distinguishes the physical demands of the different player groups is the distance they need to cover. The width of the outfield is almost three times as large as the width of the infield. As a result, acceleration and high-speed running ability or sprint ability over short distances is a valuable tool for an outfielder to possess. In addition, since the ball is hit off the bat with an average velocity of 86.6 mph (139.4 km/h), the outfielders have only about 2 s to get beneath a ball hit in the air, which highlights the importance of a rapid acceleration for outfield players when balls are not hit directly toward them, as usually happens. Because

of these position-specific demands, repeatable lower-extremity power is an important attribute of position players relative to pitchers and catchers. Depending on the context of a game, a position player may make multiple high-intensity efforts in the field or while on offense. However, an athlete also may participate in a game without having to produce any high-intensity efforts. Thus, an athlete must be prepared to meet the demands of a game regardless of the circumstances. HIIT is a versatile tool that can be used to build and maintain the aerobic conditioning essential for health and recovery over the course of a season, while also allowing athletes the ability to train to produce explosive power, sprint at full intensity to field their position, and recover quickly between efforts. In addition, HIIT can be applied to manage training load and neuromuscular strain as needed throughout the course of a season.

Current research into the physical attributes of infielders suggests that middle infielders (2B and SS) have similar pro agility scores (4.38 ± 0.15 s versus 4.35 ± 0.19 s, respectively) as outfielders. However, both of these groups are quicker than corner infielders (1B and 3B; 4.53 ± 0.14 s) (9). Interestingly, 10 yd (9.1 m) sprint performance was statistically faster in outfielders versus corner infielders (1.5 ± 0.1 s versus 1.6 ± 0.1 s, respectively), while middle infielders exhibited a value between these two groups. These performance times have been shown to improve with the levels of professional baseball league standards in the United States (5).

Key Weapons, Manipulations, and Surveillance Tools

This section presents a sample of the different HIIT weapons used in baseball, their manipulations, and strategies for monitoring the effectiveness of training. There exists a paucity of research literature discussing conditioning strategies for baseball performance. One issue driving this is the lack of consensus among practitioners and researchers as to the most important physical characteristics and effective training strategies needed to enhance baseball-specific performance. Traditionally, a primary conditioning strategy for pitchers has been long, slow distance training (12). Several factors made this form of training popular, including the lack of need for equipment, minimal risk of acute injury, anecdotal support of the approach by successful pitchers,

and an erroneous belief that lactic acid accumulation was the source of post-pitching fatigue and soreness, with long runs used as a means of “flushing” the acid from the system. A parallel situation is also described in the ice hockey chapter, whose players share similar physical demands (figure 22.3).

Over the past 10 to 15 yr, a better understanding of the physical characteristics needed to be a successful baseball player has emerged. In turn, strategies to effectively develop these characteristics have improved. As outlined, baseball requires quick, explosive movements and repetitive expression of instantaneous power. As enhanced methods of energy system training to facilitate these characteristics have evolved, HIIT has emerged as an integral component of baseball conditioning programs. A study by Rhea et al. (10) showed significant improvements in power in college baseball players engaging in HIIT versus a group performing submaximal cardiovascular endurance training. In addition to power production, HIIT enhances skeletal muscle oxidative capacity to support cellular energetic recovery between efforts (3). Today HIIT is considered a ubiquitous component of conditioning programs throughout the training calendar for baseball players with the weapons formed in accordance to the position-related requirements and specific individual needs.

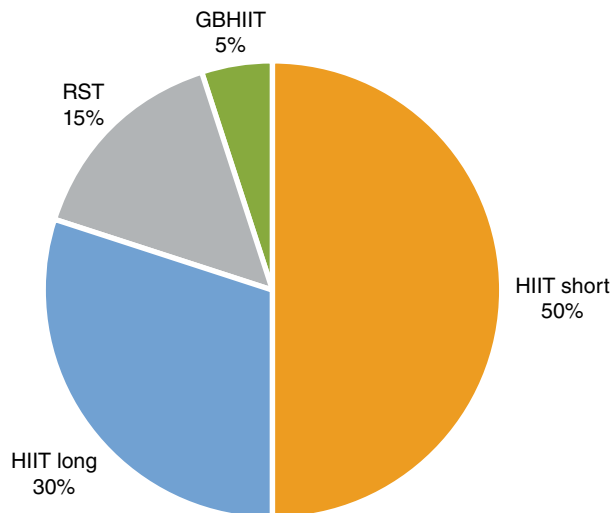


Figure 22.4 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite baseball. HIIT: high-intensity interval training; RST: repeated-sprint training; GBHIIT: game-based HIIT.

HIIT Weapons

The HIIT weapons used to condition baseball players target all response types 1 to 5, as described in chapter 5, and are used at different times throughout the year, based on the time of the season, positional demands, and the individual characteristics of the player (figure 1.5). HIIT session modalities can include straight-line sprinting, sprints with changes of direction, shuttle runs, plyometric exercises, resistance-based circuit training, or conditioning exercises that use indoor stationary training equipment such as a cycle ergometer, indoor rowing machine, or VersaClimber. The very large majority of HIIT includes short intervals, followed by long intervals and repeated-sprint training (figure 22.4).

Manipulation of Interval Training Variables

HIIT programming for baseball players is systematically structured in an attempt to reach the desired acute metabolic and neuromuscular responses (i.e., physiological target types 1, 2, 3, 4, or 5), to condition the athlete appropriately throughout the season. Conditioning, just like strength training, needs to be progressed or regressed to match the desired metabolic training or recovery effect, while also providing an appropriate level of neuromuscular strain on the athlete.

Before training variables can be manipulated, a baseline level of physical performance needs to be established with each athlete, prior to routine serial testing, to ensure that the program is meeting the needs of the athlete. In some instances, testing may be performed to ensure that an athlete possesses a baseline level of a specific attribute. For example, athletes may be asked to run a certain distance in a given amount of time. They may have to repeat several efforts at this distance with established rest intervals. If an athlete is unable to satisfactorily achieve a certain standard in a test, the program will be modified to remedy the deficient attribute.

Testing might also be structured to examine an athlete's maximal capabilities. For example, an athlete may perform an activity for a set time with the instruction to produce the maximum output possible during this period. Training variables such as volume and intensity may then be modified according to this result. This allows for more precise structuring of conditioning intervals and ensures that each athlete in the team setting is given an individualized con-

conditioning prescription, as opposed to all members of the team performing the same program with the same parameters.

Once testing is completed and a training program is established, several key variables will need to be manipulated throughout the training program. Key variables we manipulate include the training modality, intensity, work/rest intervals, and volume (see also chapter 4). Each variable is structured relative to several factors, including, but not limited to, the time of year, athlete age, anthropometry, body composition, playing position, health, and workload status.

Baseball training most often follows a relatively standard periodization schedule, with a competitive phase (CP), transition phase (TP) following the season, general physical preparation phase (GPP), and specific preparation phase (SPP). For HIIT, much like traditional strength training, it is leveraged differently during each phase of the year.

Age is an important factor to consider in the manipulation of training variables, as professional baseball players range from teenagers to individuals in their late thirties and even early forties. This highlights the importance of addressing any unique needs based on an individual's stage of development, training age, physical maturity, and injury history. As discussed by Mangine et al. (8), players who continue their careers beyond age 35 must demonstrate the ability to maintain similar levels of speed, agility, and power as they possessed earlier in their careers. The conditioning program for these athletes plays an integral role in their career longevity and should be structured with these goals in mind.

Conversely, young baseball players are often deficient in the physical attributes necessary for success at the highest level. These athletes' programs should reflect the need for systematic progression of strength, speed, and power development. In addition, the enhanced recovery ability of a young athlete means that shorter recovery times may be leveraged, although appropriate recovery needed within and between training sessions should always be assessed on an individual basis.

Body type and composition are also important factors in determining HIIT programming, with norms across positions already outlined (9, 11). HIIT can be a key tool in helping achieve body mass and composition goals, when structured appropriately. In many instances, young athletes have difficulty maintaining body mass throughout a season. Games are typically played in warm weather and the volume

of games, combined with varying dietary options, can make it difficult for athletes to consistently receive adequate nourishment (chapter 7). Conversely, some athletes have difficulty maintaining appropriate body composition during a season due to inconsistency of workload. In the off-season, strength and conditioning and nutrition are much easier to control and monitor. During the season, if an athlete is unable to maintain consistency in his conditioning and nutrition programs, body composition can reach suboptimal levels that can affect performance.

Workload is another variable that is difficult to predict and must be addressed. Throughout the season, athletes are subjected to highly variable workloads based on game schedules, individual playing time, and the nature of each competition. Athletes may be well recovered for training or may be dealing with fatigue or issues with joint and soft tissue health. HIIT must be performed in a manner so that the musculoskeletal health of the athlete is not compromised. Use of HIIT target types 1 and 3 can be an effective strategy for mitigating neuromuscular load while allowing the athlete to derive the intended metabolic training effect.

Short-Interval HIIT

As described earlier, baseball is a sport that in most instances requires some combination of acceleration, change of direction, explosive power, and reactive strength to succeed (figure 22.3). The dimensions of a baseball diamond (figure 22.1) limit the distance a player may sprint to typically no more than 270 ft (82.3 m) and more routinely 90 to 180 ft (27.4 to 54.9 m). Thus, short-interval HIIT is an integral component of baseball conditioning throughout the year.

During the general preparation phase (GPP), athletes typically use short-interval training 1 or 2 times per wk. Athletes who have an offseason training program designed to add muscle mass and strength will typically perform this type of conditioning more frequently. Athletes who have a goal of fat loss instead would be prescribed more long-interval HIIT as a foundational component of their program.

Modalities used for short-interval HIIT during GPP are straight-line sprints (types 1 and 3), shuttle runs (type 2), indoor stationary equipment (primarily type 1, e.g., treadmill, bike, VersaClimber), and body-weight strength training exercises (type 4). Work intervals are typically 10 to 45 s, with a work:rest ratio ranging from 2:1 to 1:4. Intensity is typically prescribed with a suggested rating of perceived exertion (RPE) of 0 to 10 (0 = nothing, 10 = very,

very heavy intensity; chapter 8), with proportionately higher intensities prescribed to shorter work or longer rest intervals.

As the program transitions to the baseball-SPP, a majority of HIIT involves straight-line runs (type 1) and runs with changes of direction (type 2). Change-of-direction sprinting will typically involve a 90° turn, roughly simulating the act of running bases.

During the competition phase, athletes are often observed sprinting before games as part of their warm-up routine, as well as during the game itself. Managing an athlete's load in this regard is a challenge, and it can prove difficult to compress training into a schedule that includes 5 to 7 games per wk. Thus, decreasing neuromuscular strain and mechanical load on the body is important, using predominantly type 1 (and to a lesser extent, type 3) HIIT targets. For that reason, it is common during the season to use modalities such as a stationary bike, VersaClimber, or rowing ergometer, with each modality offering individual advantages and varying levels of strain and load (see chapter 8).

Depending on the athlete, it is also possible to use reactive agility training (type 4) as a conditioning tool, with quick changes of direction in response to varied stimuli. This would be most appropriate with an athlete who has not participated in game action recently and would benefit from the increased load of the activity.

During the transition phase following the season, short-interval HIIT is incorporated in a fashion similar to its utilization during the competition phase. The goal of the transition phase is for the athlete to recover from the stress of the season, so shorter intervals using low-impact modalities (type 1) serve well in this capacity. The transition phase is typically short and should allow the athlete to prepare for the enhanced volume and intensity of the general preparation phase.

Long-Interval HIIT

Long-interval HIIT is most commonly used during the general preparation phase as a means of building aerobic capacity. During the competitive season, long-interval HIIT is leveraged less frequently, but is still used with most athletes. In particular, pitchers will often perform long-interval training throughout their recovery period between pitching appearances. High aerobic capacity has been shown to be an important attribute of starting pitchers, and long-interval HIIT supports this training goal (4). In addition, athletes who have

not appeared in a game over several days may need to maintain work capacity through use of long-interval training.

The most common modalities used for long-interval HIIT are shuttle runs (type 4) and stationary cycling (type 1). These modalities are often chosen because they impart no load on the upper extremity of the throwing athlete. In addition, equipment and space is often readily available in most baseball venues. If upper-extremity load is desired, modalities such as a rower and VersaClimber can be considered.

Often during the competitive phase, strategies and training devices are incorporated that increase work performed by local muscles while decreasing the time and impact necessary to achieve the desired training effect. One example is the use of personalized blood flow restriction devices, which limit specific amounts of arterial and venous flow into and out of the limbs. The altered blood flow pattern induces local and systemic responses that produce a training effect with less training volume (6). Intervals last typically 1 to 5 min, with work:rest ratios of 2:1 to 1:2.

An additional HIIT format common throughout baseball is to use the warning track of the outfield as a running surface. Athletes will run from one foul pole to the other, with different prescriptions for distance covered and recovery intervals based on the training context and recovery variables necessary at the time of the training session. This is historically a less-structured form of training, as intensity is often self-selected and therefore can be highly variable.

Repeated-Sprint Training

RST is an important component of baseball conditioning, particularly during the general preparation and specific preparation phases. During the competitive and transition phases, repeated sprints (type 4) are less frequently used, as a primary goal is to minimize neuromuscular load and overall training volume. In addition, preservation of soft tissue health is paramount during the season, and RST carries an increased risk of acute injury.

The structure of RST during the GPP and SPP phases typically involves straight-line sprinting for distances such as 30 or 60 yd (27.4 to 54.9 m), with passive recovery periods typically ranging from 20 to 45 s, depending on the distance covered in the sprint. Volume of training is structured relative to the time of year and other training activities. Typically, athletes will complete 1 to 2 sets of 10 to 12 repetitions, with 4 to 6 min of rest between sets.

If utilized during the competitive or transition phases, volume is at the discretion of the coach. The coach must determine how much volume is necessary to provide the training effect the athlete needs based on his level of competitive activity. Most typically, this will include fewer repetitions and greater rest intervals between efforts. A common prescription of an RST session for a baseball player might be 6-8×30 yd (27.4 m) sprints with 45 to 60 s of rest between efforts.

Game-Based High-Intensity Interval Training

Game-based high-intensity interval training (GBHIIT) is not used as often in baseball conditioning as it is in other team sports. The nature of the game of baseball, requiring pitchers to throw at high velocities, does not lend itself to an efficient GBHIIT approach. However, activities during baseball skill practice, such as fielding outfield or infield, running the bases, and taking batting practice, often involves some degree of interval efforts. Coaches may guide athletes through situational drills, in which simulated game events are practiced in the field and while running bases. The lack of structure and predictability in this type of activity makes it difficult to classify it as any other specific type of HIIT other than GBHIIT. Examples of GBHIIT in baseball might be working on double-play routines at game speed using four groups of two middle infielders at a 1:3 work:rest ratio or working on situational baserunning during which baserunners respond to various types of hits (singles, doubles, etc.) in a simulated setting as runners cycle through the drills at appropriate work-rest intervals.

Surveillance and Monitoring Tools

In-season monitoring tools used in professional baseball are team specific and few common practices have yet to be established. A majority of player surveillance models have focused on measuring shoulder motion consistently due to the available research associating these measures with injury risk for pitchers. In looking at the position-specific needs analysis, it seems appropriate to monitor measures of upper-extremity power, lower-extremity power, rotational power, lower-extremity endurance, linear acceleration, and lateral agility. Currently, skill-based workloads are examined by charting the number of pitches thrown in a game or the number of swings taken during batting practice. While the monitoring of skill-specific workloads is beneficial, they should

be taken into context of the overall workload in order to understand if they have a meaningful effect on overuse injuries across a 6 mo season or have a deleterious effect on skill development.

Based on the game length and load density of professional baseball, questioning a player's fatigue may be of benefit in identifying a player's response to training workloads (chapters 8 and 9). Combining the physical preparation plan, alongside skill-specific training workload data and subjective workload reports, may offer the practitioner a holistic view of the training load-input response cycle in order to systematically identify how a player should modify future training workloads in order to maximize long-term development.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in baseball are influenced by various controllable and uncontrollable factors. This section discusses some of the nuances regarding the integration of HIIT into the strength and conditioning program for baseball.

Controllable and Uncontrollable Factors

The schedule throughout a baseball season presents logistical challenges to maintaining a consistent strength and conditioning program. Throughout the year, some factors are consistent and relatively controllable, whereas other factors are highly variable and difficult to predict. During various phases of the off-season (GPP, SSP, TP), most training factors are controllable. Athletes are typically on predictable and modifiable skill training programs, and strength and conditioning work is of high priority during this period to build resiliency for the upcoming season. Athletes often reside in one location during the off-season and can maintain a consistent training schedule with access to necessary facilities, equipment, and coaching.

Based on the predictable nature of the offseason as opposed to the highly variable schedule during the competitive season, most significant gains in strength and conditioning metrics are achieved during the offseason training cycles. It is also more acceptable for an athlete to experience appropriate levels of soreness and fatigue from training during off-season phases, though athletes are still closely monitored and health remains the highest priority.

During the competitive season, workload and schedule present the most significant obstacles to training. Workloads for players can vary considerably from week to week and from position to position. Starting pitchers may pitch on a regular schedule, while relief pitchers' appearances may be highly variable from week to week. Starting pitchers still typically perform supplemental conditioning work throughout the season. Some of the planning is contingent on the pitcher's preference, as certain athletes desire more consistent conditioning sessions between competitive events. Relief pitchers, on the one hand, might pitch across multiple days in a row or may not pitch for several days in a row. In addition, a pitcher might throw 1 pitch or he might throw 120 pitches. The variability in workload must be met with a flexible conditioning schedule that mirrors the needs of the athlete.

Position players may also have inconsistent workloads from day to day. Some athletes are in the lineup almost daily, while other players may wait several days between game appearances. In addition, position player workloads are highly dependent on the context of the game. A position player's load is directly correlated with the number of sprint efforts he makes during the game, both in the field and on the bases. Player conditioning programs must therefore be modified to reflect the volume of sprinting performed in the days preceding the training session.

Travel presents an additional challenge during the competitive season. Athletes may be on the road (hotels, airports, and buses) for 2 wk at a time, with cross-country flights occurring several times throughout the year. The fatigue from late-night travel, shifting time zones, and altered sleep schedules can leave the athlete feeling too fatigued to follow a normal

Table 22.1 Sample Training Program for a Position Player With HIIT Targets

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
GPP	Lower body+short-interval HIIT (types 1, 3, 4)	Upper body	Mobility work	Lower body+long-interval HIIT (type 1 or 4)	Upper body	Short- or long-interval HIIT (types 1, 3, 4)	Recovery
SPP	Lower body+RST (type 4)	Upper body	Mobility work	Lower body+short-interval HIIT (types 1, 3, 4)	Upper body	RST (type 4)	Recovery
CP	Game	Short HIIT (type 1)+game	Game	Full-body workout	Game	Game	Full-body workout+game
TP	Short-interval HIIT (type 1 or 3)	Full-body workout	Recovery	Short HIIT (type 1 or 3)	Full-body workout	Recovery	Recovery

GPP: general physical preparation phase; SPP: specific preparation phase; CP: competitive phase; TP: transition phase; HIIT: high-intensity interval training; RST: repeated-sprint training

Table 22.2 Sample Training Program for a Starting Pitcher With HIIT Targets

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
GPP	Lower body+short-interval HIIT (types 1, 3, 4)	Upper body	Mobility work	Lower body+long-interval HIIT (type 1 or 4)	Upper body	Short- or long-interval HIIT (types 1, 3, 4)	Recovery
SPP	Lower body+RST (type 4)	Upper body	Mobility work	Lower body+short-interval HIIT (type 1, 3, or 4)	Upper body	RST (type 4)	Recovery
CP	Game (start)	Full-body workout	Long-interval HIIT (type 1)	Recovery	Prep work	Game (start)	Full-body workout
TP	Short-interval HIIT (type 1 or 3)	Light full-body workout	Recovery	Short-interval HIIT (type 1 or 3)	Light full-body workout	Recovery	Recovery

GPP: general physical preparation phase; SPP: specific preparation phase; CP: competitive phase; TP: transition phase; HIIT: high-intensity interval training; RST: repeated-sprint training

training schedule. Facilities on the road may also offer varying levels of accessibility and training equipment selection. It is important for the program to include training options that require minimal equipment and the ability for exercises to be performed with varying amounts of time and space.

Program Structure and Progression

As previously described, baseball strength and conditioning is often divided into some form of a standard periodization schedule, with a competitive phase, transition phase, general preparation phase, and specific preparation phase. Sample training programs for position players and starting pitchers are shown in tables 22.1 and 22.2, respectively.

The primary goals during the competitive season are health, load management and recovery, and maintenance of the physical attributes trained for performance. It is beneficial for athletes to exhibit minimal soreness or fatigue from training during the season to maximize on-field performance, which highlights the importance of types 1 and 3 HIIT in the athlete's program. Strength and conditioning is typically fit into the schedule twice per week, with the structure of each session highly context driven (tables 22.1 and 22.2).

During the competitive phase, the biggest discrepancies between the programs of pitchers and position players emerge. Pitchers, particularly starters, are on a continuous 5 d cycle (table 22.2), beginning with a pitching appearance and culminating with preparation for the next game started. During the 5 d, most pitchers have a preference for specific lifting and conditioning schedules, but in general, most pitchers try to get at least one strength training session and one conditioning session in per cycle.

During the season, the program is without significant progression or advancement in training volume or intensity, and should be designed to provide

the minimal effective dose necessary to keep athletes at or near optimal levels of physical performance.

During the off-season, there is typically a deloading period (also referred to as a transition phase), which commonly lasts 2 to 3 wk. During this phase, the focus of the strength and conditioning program is on mobility, recovery, and low volume and intensity of training. Athletes are often fatigued physically and mentally from the season, so the break in training is beneficial.

Once the athlete has had ample time to rest and recover, the program moves to the general physical preparation block, with a focus on building maximum strength, speed, and aerobic capacity, while concurrently working toward mass or strength and body composition goals. Training is structured around the basic principle of progressive overload in all facets, including conditioning. Serial testing is performed to recalibrate and progress the program each month and to ensure the program is creating the desired training effects.

As the competitive season nears, the athlete transitions from general to specific physical preparation, with the incorporation of more type 2 and 4 HIIT activities. Training during this phase is structured to more closely address key physical attributes for baseball success, including power, speed, acceleration, and agility. Players should be at peak levels of physical performance as they transition into competition.

Incorporation of HIIT

As discussed throughout the chapter, HIIT is incorporated into nearly every phase of the training calendar for professional baseball players (tables 22.1 and 22.2). Practitioners must consider each of the factors and variables discussed when determining how to best leverage HIIT to help players achieve and maintain optimal levels of physical performance throughout the season.

CASE STUDY

The importance of the targeted HIIT approach highlighted in this chapter is potentially most relevantly leveraged during long-term rehabilitation stints. For the most part, longer-term injuries tend to force the athlete to focus on the next season rather than the current season. However, due to the length of the professional baseball season, some players who sustain longer-term injuries during spring training may have the opportunity to return to activity by the end of the season and into the playoffs. In these scenarios, an a priori developed continuum of key performance indicators (KPIs) can help minimize bias due to external influences during the return-to-play decision-making process.

This case involved a right-handed starting pitcher who underwent an ACL reconstruction on his left leg (lead leg) in March. The injury occurred during spring training to which the player had reported in good physical condition. As a result, the goal was to maintain as much strength and conditioning of his body (upper-extremity and core focus due to no limitations) while his knee was given time to heal and recover from the surgery. The team had no expectation that he would return during the year.

During the early part of the rehabilitation, the focus was on restoring function to the left knee, however, equal attention was paid to make sure the rest of the body would be ready to progress with loading when the knee was ready. As with most ACL injuries, restoring knee extension precedes restoring full knee flexion, and this was the same with this surgery as well. This a priori established protocol provided some specific guidelines regarding movements that could be loaded (neutral knee extension) as opposed to movements that should resist excessive loading (full knee flexion). Priorities were established with hinging and single-leg stance movements to load as well as to utilize for training various metabolic pathways since these movements had no isolated joint limitation and remained competent with body weight load. As mentioned earlier in the chapter, a strong metabolic base is important for starting pitchers who need to have endurance as well as intermittent repeatable bouts of total-body power when pitching in the game. These movements and positions were used to identify methods to utilize HIIT to develop anaerobic power as well as maintain an aerobic fitness base.

In this specific case, the player's knee flexion did not respond well. He experienced soreness and muscle guarding with heavy sprinting or squatting. As a result, training expectations were adjusted to target sled pulls and goblet squats with heavy kettlebells to allow the incorporation of similar loading patterns in a tissue tolerant loading environment. Due to this revision in the training plan, there was minimal formal running. However, in order to efficiently test his fitness levels, it was required for him to run two 300 yd (274.3 m) shuttles (50 yd $\times$ 6) (45.7 m $\times$ 6) with a 3 min rest period between bouts. The player also completed a 5-10-5 agility test to indicate how his knee functioned under higher-level agility tasks. The times for the 300 yd (274.3 m) shuttle were a strong 53 and 51 s, respectively, and his time for the 5-10-5 would have placed him among the top performers at that year's professional football combine. While the activities were not specifically tested, it is believed that the HIIT programming likely established a solid foundation for high-level performance under these conditions. At the end of the season (5 mo since the surgery), the player was able to return to function as a starting pitcher with his team and remained in this role throughout the playoffs. This outcome was a collaboration of the player's hard work and grit as well as the staff's ability to identify multiple training pathways to achieve the relevant KPIs as efficiently as possible in order to support the players on-field goals.